Nguyen Dai Dong

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dong-nguyen-hcmut   Noname-29-lnin

Education

Ho Chi Minh City University of Technology *Oct 2022 Present ReNov 2026*
Bachelor of Science in Electronics Engineering

- GPA: 3.0/4.0 (This link)
- **Project 1:** Design and Implementation Viterbi Decoder Algorithm on FPGA.

Research Interest

Computer Architecture, RISC-V Processors, Hardware Accelerator, Digital IC Design.

Technical Skill

- Programming Language & HDL:
  - Proficient: C, C++, SystemVerilog, Verilog.
  - Scripting & Documentation: LaTeX, Bash.
  - Tools: Vim, Git, Makefile.
- Development Environments:
  - Operating Systems: Linux (Ubuntu, CentOS), Windows.
  - FPGA Tools: Intel Quartus, QuestaSim, ModelSim, Verilator.
- Hardware Skills:
  - Digital Design: RTL development, Verification.
  - PCB Design: KiCad, Altium (Basic).
  - Lab Equipment: Oscilloscope, Logic Analyzers.

Projects

Design and Implementation of Viterbi Encoding and Decoding Algorithm on FPGA *Noname-29-lnin /Viterbi_Decoding*

- This project involves the hardware design and implementation of a Viterbi decoder for convolutional codes with parameters (3, 1, 2) (constraint length K=3, code rate 1/2) on an FPGA. The Viterbi algorithm is optimized for error correction in noisy communication channels, leveraging trellis-based path metrics and survivor path selection.
- Tool & Languages:
  - Simulation: Verilator.
  - Synthesis: Intel Quartus Prime.
  - Languages: C, SystemVerilog.
  - Build System: Makefile.

Design of a Single Cycle RISC-V Processor *Noname-29-lnin /Single-Cycle-RISC-V*

- A single-cycle implementation of the RISC-V RV32I instruction set architecture (ISA), designed from scratch in SystemVerilog. This project demonstrates the core principles of CPU microarchitecture, including instruction fetch, decode, execution, memory access, and writeback stages—all completed within a single clock cycle.
- Tool & Languages:
  - Simulation: Verilator, Xcelium.

- Synthesis: Intel Quartus Prime.
- Languages: SystemVerilog.
- Build System: Makefile.

Implementation Adder 32bit using Verilog *Noname-29-lnin/Adder_32bit*

- A comprehensive implementation and comparison of four classic 32-bit adder architectures in Verilog, analyzing the trade-offs between speed, area, and power consumption. Implementation Architecture:
  - Carry Ripple Adder (CRA)
  - Carry Lookahead Adder (CLA)
  - Carry Select Adder (CSA)
  - Carry Skip Adder (CSKA)
- Tool & Languages:
  - Simulation: Verilator.
  - Synthesis: Intel Quartus Prime.
  - Languages: C++, Verilog.
  - Build System: Makefile.

Design basic Vending Machine *Noname-29-lnin/Vending_machine*

- A hardware implementation of a finite state machine (FSM) based vending machine controller in Verilog, simulating the complete transaction flow from item selection to product dispensing.
- Tool & Languages:
  - Simulation: Verilator.
  - Synthesis: Intel Quartus Prime.
  - Languages: C++, Verilog.
  - Build System: Makefile.

Design of a DC/AC Voltage Measurement and Waveform Display Device Using STM32F1xx *Noname-29-lnin/OSC_project*

- An embedded measurement system using STM32F1xx microcontroller for accurate voltage acquisition and real-time waveform display on PC, implementing core functionalities of a basic oscilloscope.
- Tools Used: KiCad, Proteus 8.
- Language: C, LaTex.
- Tool & Languages:
  - Hardware Design: KiCad.
  - Simulation: Proteus 8.
  - Languages: C.
  - Development Tools: STM32CubeMX, Keil-C, ST-Link programmer.